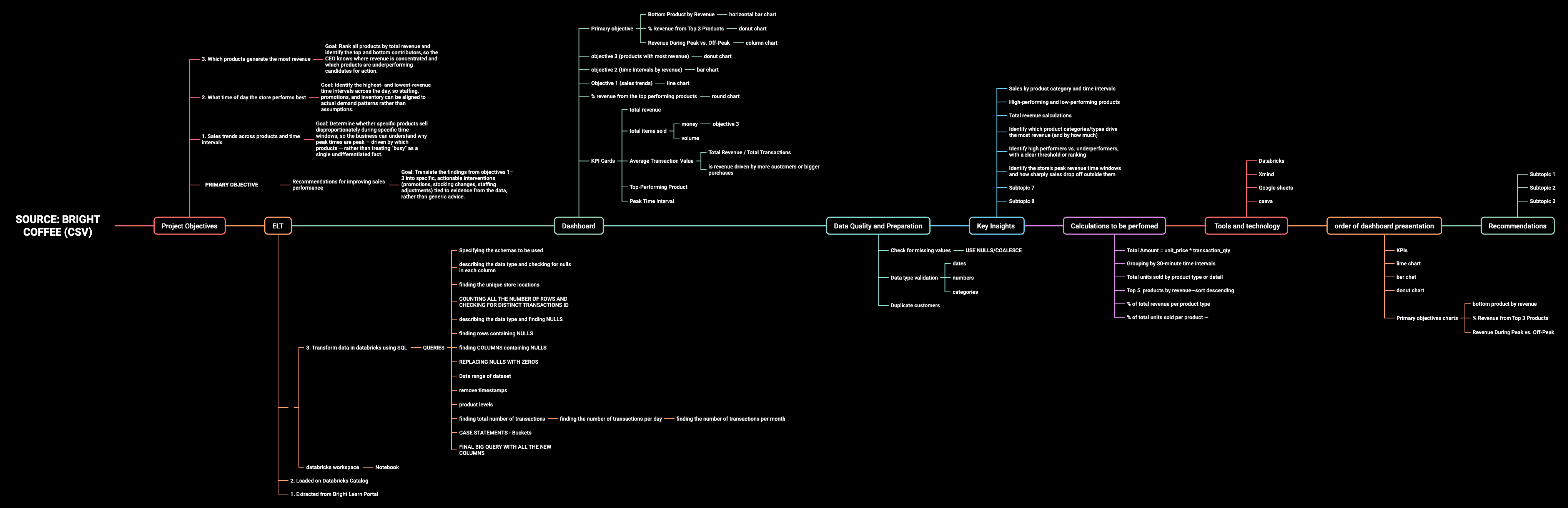

SOURCE: BRIGHT COFFEE (CSV)

- **Project Objectives**
  - 3. Which products generate the most revenue
    - Goal: Rank all products by total revenue and identify the top and bottom contributors, so the CEO knows where revenue is concentrated and which products are underperforming candidates for action.
  - 2. What time of day the store performs best
    - Goal: identify the highest- and lowest-revenue time intervals across the day, so staffing, promotions, and inventory can be aligned to actual demand patterns rather than assumptions.
  - 1. Sales trends across products and time intervals
    - Goal: Determine whether specific products sell disproportionately during specific time windows, so the business can understand why peak times are peak — driven by which products — rather than treating "busy" as a single undifferentiated fact.
  - PRIMARY OBJECTIVE
    - Recommendations for improving sales performance
      - Goal: Translate the findings from objectives 1–3 into specific, actionable interventions (promotions, stocking changes, staffing adjustments) tied to evidence from the data, rather than generic advice.
- **ELT**
  -
    - 3. Transform data in databricks using SQL
      - QUERIES
        - Specifying the schemas to be used
        - describing the data type and checking for nulls in each column
        - finding the unique store locations
        - COUNTING ALL THE NUMBER OF ROWS AND CHECKING FOR DISTINCT TRANSACTIONS ID
        - describing the data type and finding NULLS
        - finding rows containing NULLS
        - finding COLUMNS containing NULLS
        - REPLACING NULLS WITH ZEROS
        - Data range of dataset
        - remove timestamps
        - product levels
        - finding total number of transactions
          - finding the number of transactions per day
            - finding the number of transactions per month
        - CASE STATEMENTS - Buckets
        - FINAL BIG QUERY WITH ALL THE NEW COLUMNS
    - databricks workspace
      - Notebook
  - 2. Loaded on Databricks Catalog
  - 1. Extracted from Bright Learn Portal
- **Dashboard**
  - Primary objective
    - Bottom Product by Revenue
      - horizontal bar chart
    - % Revenue from Top 3 Products
      - donut chart
    - Revenue During Peak vs. Off-Peak
      - column chart
  - objective 3 (products with most revenue)
    - donut chart
  - objective 2 (time intervals by revenue)
    - bar chart
  - Objective 1 (sales trends)
    - line chart
  - % revenue from the top performing products
    - round chart
  - KPI Cards
    - total revenue
    - total items sold
      - money
        - objective 3
      - volume
    - Average Transaction Value
      - Total Revenue / Total Transactions
      - is revenue driven by more customers or bigger purchases
    - Top-Performing Product
    - Peak Time Interval
- **Data Quality and Preparation**
  - Check for missing values
    - USE NULLS/COALESCE
  - Data type validation
    - dates
    - numbers
    - categories
  - Duplicate customers
- **Key Insights**
  - Sales by product category and time intervals
  - High-performing and low-performing products
  - Total revenue calculations
  - Identify which product categories/types drive the most revenue (and by how much)
  - Identify high performers vs. underperformers, with a clear threshold or ranking
  - Identify the store's peak revenue time windows and how sharply sales drop off outside them
  - Subtopic 7
  - Subtopic 8
- **Calculations to be perfomed**
  - Total Amount = unit_price * transaction_qty
  - Grouping by 30-minute time intervals
  - Total units sold by product type or detail
  - Top 5 products by revenue—sort descending
  - % of total revenue per product type
  - % of total units sold per product —
- **Tools and technology**
  - Databricks
  - Xmind
  - Google sheets
  - canva
- **order of dashboard presentation**
  - KPIs
  - lime chart
  - bar chat
  - donut chart
  - Primary objectives charts
    - bottom product by revenue
    - % Revenue from Top 3 Products
    - Revenue During Peak vs. Off-Peak
- **Recommendations**
  - Subtopic 1
  - Subtopic 2
  - Subtopic 3

Presented with xmind

Legend
Insert markers into the topic